Zoology Legends

BIO201

Current Final TERM Spring 2023

Muhammad Zaman

0304-4756496

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Zoology Legends

MCQS Spring 2023

→ Inter membrane pores for water are called...**aquaporins**

→ Function of G protein... **transmit signals form cells**

→ Mendel work on which plant...**pasivum sativum (PEA)**

→ Power house organelles... **Mitochondria**

→ Function of ER... **protein synthesis**

→ When DNA damage which protein activate...**p53 intiate cellular responses**

→ **Somatic mutations:** Occur in somatic (body) cells & passed on to the daughter cells after mitosis but not passed on to the children.

→ Mast cells Release **histamine** when damaged

→ Histamine release from **mast cells** vascular permeability

→ All proteins synthesis initiates in the **cytosol, (mitochondria & plastids),**

→ tRNA Translates Nucleotide Language **into AA Language**

→ _ is removed during RNA maturation? **Intron**

→ activators consist how many domains? **Multiple domains**

→ what charge is on DNA? **Negative**

→ What will be mixture of DNA and proteins? **Chromatin**

→ Point mutation has base pair? **Single**

→ smad complex moved towards dash? **Nucleus**

Zoology Legends

→ There are **2×10^{12} lymphocytes** in the human body, making the immune system comparable in cell mass to the liver or brain.

→ The B-Cells It is composed of **four polypeptide** chains 2 identical heavy chains & 2 identical light chains.

→ Autonomous Cell Growth **In Cancer Cells**

→ nucleus discovered by .. **robert Brown**

→ Human blood group discovered by ....**karl landsteiner**

→ Cluster of genes with ------ promoter is called **operon**? **1**

→ Mutation that cause no change .... **silent mutation**

→ Inhibitors that binds to a site other than active site of enzymes---- **Non-competitive inhibitors**

→ MYC and the **polymerase** assemble on a specific DNA sequence and drive transcription of genes.

→ Condition in which cells have specific binding**. Cell adhesion**

→ Cyclin D and Cycline E produced in? **G1 phase**

→ An average cell can divide **50 times** before it gets unhealthy

→ CML results from a chromosomal translocation between chromosome **9 and 22**

→ Housekeeping genes are responsible for the routine metabolic functions **(e.g. respiration).**

→ TH1 cells mainly help activate **macrophages and cytotoxic T cells**

→ TH2 cells mainly help **activate B cells**

Zoology Legends

→ Prolactin stimulates breast cells to produce **milk**

→ After 30 cycles almost ----- number of copies are formed. **16 billion**

→ Lipids are....**non soluble**

→ Skin regenerate every -------? **Two weeks**

→ T cells mature in ----? **Thymus**

→ Helices that are involved in against binding are __**Helix 5**

→ Beta blockers are involved in ___**mimicking the function of a hormone.**

→ DNA is highly ordered around positively charged -----? **Histone proteins**

→ PtK stands for ___**Protein tyrosine kinases**

→ Prosthetic group example is.....**copper**

→ Unit of genetic map.....**centimorgon**

→ Mitosis sequence..... **Prophase to anaphase**

→ May 2014 the media reported on a blood transfusion that Contains a protein called__**GDF11** to possibly cure Alzheimer's disease and heart Disease.

→ Even in the absence of antigen humans can make More than 10^{12} Different antibody molecules.

→ Histamine diffuses into ----? **Capillaries**

→ Mutation of RAS is present approximately -----in all humans? **30%**

Zoology Legends

→ Oncogenes after mutation produce one of abnormal condition.__ **Cancer**

→ Philadelphia chromosome is a chromosomal translocation between chromosome ----And ---- **9 & 22**

→ Smallest unit of life …**cell**..

→ For optimum activity of hydrolytic enzymes, lysosomes maintain a pH about…**4 to 5** .

→ Cluster of genes with --**1**-- promoter is called **operon**____

→ The first transcription factor is…**glucocorticoids**…

→ Which cells release histamine when damaged**…mast cells**

→ How many types of MHC molecules? **2**

→ Autonomous cell growth is present in ----- cell. **Cancer**

→ Lysosome contains pH ----? **5.0**

→ ERBB/HER2 is amplified in the ------ of the cancer? **25%-30%**

→ TH2 cells mainly help activate **-------? B cells**

→ **5 classes** of antibodies, IgA, IgD, IgE, IgG, & IgM, class of heavy chain—α, į, İ, γ, & μ, respectively.

→ **Four human IgG** subclasses (IgG1, IgG2, IgG3, and IgG4), having γ1, γ2, γ3, andγ4 heavy chains, respectively.

→ Organisms shuffle genes by **Meiosis**

→ Which mutation passes on the daughter cells after the mitosis? **Somatic mutation**

→ UUU code for…. **Amino acid phenlyalanine**

Zoology Legends

→ T cell meture in ....**thymus**

→ **Antagonists (β-blockers):** Bind to the receptor but do not activate hormone induced effects.

→ **Stem cells** are unspecialized cells that renew themselves.

→ Which cells don't cause e rejection? **Adult stem cells**

→ Genes are located in the nucleus and tangled chromatin material wrapped around histone in----**Eukaryotes**

→ DNA replication obeys....**Semi- conservative model**

→ Difference between the alpha 1-4 glycosidic and beta 1-4 glycosidic bonds is on the basis of .....**Size and Shape**

→ Which one of the following helps in stimulating breast cells to produce milk? **Prolactin**

→ Ribosomes are engaged in ---- **protein synthesis**

→ Even in the absence of antigen humans can make more than.....different antibody molecules. **10^{12}**

→ Initiation site of transcription... **Promoter**

→ Ran GTpase is a molecular switch that exits in .....**2 states**

→ The lipids molecules held together by weak interactions and ----- **Van der Waals forces**

→ **During R**NA maturation are Removed...**introns**

→ The small subunits of ribosomes are bind to mRNA this sequence is called? **Shine Dalgarno sequence**

→ TH2 cells mainly help activate -------**? B cells**

Zoology Legends

→ Which is the first class of Ab to appear on the cells surface? **lgM**

→ How many years ago life was originated **4,000,000,000**

→ **TATA** box where DNA begins to denature?

→ Erythropoietin stimulates erythrocyte precursor cells to survive, proliferate, and differentiate, is associated with **Jak2.**

→ T cells mature in... **Thymus**

→ .....is the structural and functional unit of life.. **cell**

→ In the abscense of antigens....**10^12** tha

→ Milk production by.....**prolactin**

→ oncogenes causes .....**cancer**

→ T-site on large subunit is for....**transcription,translation etc**

→ Helper T cells activated by an ---- cell to become an affecter cell? **Antigen cell**

→ Which of these act as phagocytes.....**macrophage, dendritic or leuphilles all**

→ Positive Control of Transcription the regulatory molecule (**the CRP–cAMP** complex) enhances transcription

→ which molecule can convert light energy.... **NAD**

→ simian crease is .....**single Palmer fold**

→ genetic Information from DNA is translated through..... **mRNA**

→ proteins with promoter is called.. **transcription factor**

→ Value of free energy at equilibrium.... **zero**

→ DNA forming nucleosome is found in.....**eukaryotes**

Zoology Legends

→ TH2 help to activate.... **B cells**

→ human body can make antibodies in the absence of antigens.**....10 power 12**

→ the number and appearance of chromosome.... **Karyotype**

→ the substance in the immune system causes capillaries to dilate.... **Histamine**

→ Philadelphia is a chromosomal translocation between chromosome**...9 & 22**

→ dendritic cells use following signal to activate r cells...**. IL-2**

→ ..... Can be removed after translation... **Interns**

→ proteins which are fully translocated along mitochondria are.... **Water soluble proteins**

→ smad complex move into ..... **Nucleus**

→ TH2 cells mainly help to activate? **B cells**

→ The .... immune system, like the ... system can remember prior experiences? **Adaptive, nervous**

→ The fusion of membranes from same compartment is termed as? **Homotopy fusion**

→ For optimal activity of hydrolytic enzymes lysosomal maintains pH about?? **5.0**

→ Which one of the following substances in immune system causes capillaries to dilate and become leaky? **Histamine**

→ Ran GTPase is a molecular switch that exist in? **2 states**

Zoology Legends

→ An oncogene after mutation produces one of abnormal condition? **Cancer**

→ How many times an average cell divide before it gets unhealthy? **50 times**

→ Philadelphia chromosome is a chromosomal translocation between? **9&22**

→ Genes are located in the nucleus and tangled chromatin material wrapped around histone is ----**Eukaryotes**

→ Condition in which cells have specific binding... **Cell adhesion**

→ The fusion of membranes from the same compartments is termed as ---- **Homontypic fusion**

→ Proteins which are fully translocated across Endoplasmic reticulum and are embedded in it ------ **Water soluble proteins**

→ Which one of the following helps in stimulating breast cells to produce milk? (**Prolactin**)

→ Philadelphia chromosome is a chromosomal translocation between chromosome ---**(9 & 22)**

→ GTpase is a molecular switch that exits in **(2 states)**

→ Dendritic cells use on which of the following signal to guide helper T cells to effector T cells **--(IL- 2)**

→ Mutation of RAS is present approximately -----in all humans? **(30%)**

→ RAN GTpase is a molecular switch that exists in__ states: **2 states**

Zoology Legends

→ TH 2 cells helps to activate : **B cells**

→ Identify the initiation codon **AUG**

→ The protein structure is :**Three Dimensional 3D**

→ **IgG** only Ab that can pass from mother to fetus placenta cells

→ MHC can accommodate an extended peptide about ----- long? **8-10 amino acid**

→ TH2 cells mainly help activate -------? **B cells**

→ Lysosomes has PH ___? **5.0**

→ Mutation of __ is present approximately **30%** in human? ***RAS***

→ For optimum activity of hydrolytic enzymes, lysosomes maintain a pH about**. 5.0**

→ Beta blockers are involved in **mimicking the function of a hormone.**

→ An NLS is necessary for import of proteins into the ------? **Nucleus**

→ TATA box where ---- begins to denature? **DNA**

→ Function of Golgi apparatus is ------ **Packing and transportation of proteins**

→ Transfer of materials was from cytosol to **ER**

→ Number of macromolecules in cell?_**4**

→ Calmodulin is about..... % of total cell protein mass**?** **10%**

→ How many protein molecules an animal cell contains ? **10 Billion**

Zoology Legends

→ Some of the genes those who remain expressed all the time in cells are called? **Constitutively Expressed Genes**

→ --------- Energy is the energy of motion_**Kinetic**

→ Point mutation has base pair? **Single**

→ Redundant in codon means----- **One codon specify more than one amino acids**

→ How many sub compartments of mitochondria? **2**

→ B cells Differentiate form antibody-**producing cells and memory cells**

→ Unit of genetic map..... **Centimorgan**

→ Who discover Restrictions Endonucleases..... **Arber**, **Smith & Nathans**

→ Ligand activates --- proteins? **G**

→ DNA Replicate at which stage of interphase...**.... S phase**

→ Division of cytoplasm called..... **Cytokinesis**

→ An average cell can divide ..... before it gets unhealthy. **50 Times**

→ **Stem cells** are unspecialized cells that renew themselves.

→ Bones regenerate every **10 years**

→ Skin regenerates every **two weeks**

→ Golgi apparatus found in..... **Eukaryotic Cells**

→ T-cells activate ..... **B-cells**

→ the milk production harmone is **Prolactin**

Zoology Legends

→ The binding of protein nucleotide on the surface a MHC and the peptide present near the t cell is which type of MHC___**MHC II**

→ Condition in which cells have specific binding. **Cell adhesion**

→ Missing of enzymes homogentisic acid oxidase caused -------- **Alkaptonuria**

→ Phosphorylation signaling in the E. coli alters__**Protein Synthesis**

→ Ran GTPase is a molecular switch that exist in __ **2 States**

→ Housekeeping genes are responsible for the routine metabolic **functions (e.g. respiration).**

→ T cells mature in ----? **Thymus**

→ Genes are located in the nucleus and tangled chromatin material wrapped around histone is ----**Eukaryotes**

→ Condition in which cells have specific binding... **Cell adhesion**

→ The fusion of membranes from the same compartments is termed as ---- **Homontypic fusion**

→ Proteins which are fully translocated across Endoplasmic reticulum and are embedded in it --**Water soluble proteins**

→ Which one of the following helps in stimulating breast cells to produce milk? (**Prolactin**)

→ Philadelphia chromosome is a chromosomal translocation between chromosome **---(9 & 22)**

→ GTpase is a molecular switch that exits **in (2 states)**

Zoology Legends

→ Dendritic cells use on which of the following signal to guide helper T cells to effector T cells **--(IL- 2)**

→ Mutation of RAS is present approximately -----in all humans**?** **(30%)**

→ DNA replication obeys___**Semi conservative model**

→ Enzyme that can cut DNA at specific site____**Restriction enzymes**

→ In mitosis the replicated DNA divides amongst the cell____**M phase**

→ Bacteriophage destroys__ **Bacteria**

→ which two essential sequences are present in promoter region____**TATA Box**

→ Approximately How many years ago life Originated____**4,000,000,000**

→ Helices that are involved in agonist binding are___**Helix5**

→ Beta_Blockers are involved in____**Mimicking the function of a hormone**

→ undefferentiated cells of multicellular organisms which can develope into specialized cells____**Stem cells**

→ The fusion of membranes from same compartment is termed as____**Homontypic fusion**

→ How many protein molecules an animal cell contains___**10 billion**

→ Ran GTpase is a molecular switch that exists in____**2 States**

Zoology Legends

→ Oncogenes after mutation produce one of abnormal condition____**Cancer**

→ Dendritic cells use on which of the following signal to guide helper T cells to effector T cells -------**IL_2**

→ The protein molecules which link one signaling protein to another without themselves conveying a signal are known as___**adaptor proteins**

→ Fibroblast cells can be induced to proliferate by treatment with.....**Epidermal growth factor**

→ TH2 cells mainly help to activate___ **B cells**

→ Which substance in immune system causes capillaries to dilate and become leaky? **Histamine**

→ Stimulation of milk production is due to -----? **Prolactin**

→ T cells develope in___thymus

→ In which part of cell ribosomes are synthesized? ***nucleolus***

→ Mitochondria are present in? **Cytoplasm**

→ Dynein & Kinesin are classes of__**microtubule based motor proteins**

→ Meiosis II resemble with__ **Mitosis**

→ Development of T cell takes place in____**thymus**

→ Who discovered Blood group system? **Austrian Karl Landsteiner**

→ An average globular cell have mitochondria__**10,000**

Zoology Legends

→ In 2014 it was reported that one of following proteins in blood is help fulincure of Alzheimer'sdisease and heart disease__**A12**

→ Some of the genes those who remain expressed all the time in cells are called? **Constitutive gene**

→ Sterilization also known as___**UV**

→ Function of smooth Endoplasmic reticulum___**lipds synthesizes**

→ Smooth Endoplasmic Reticulum (**SER**) is involved in lipid metabolism, detoxification, and calcium storage.

→ Factor V and Factor VIII are clotting factors in **blood coagulation**.

→ The cell was discovered by **Robert Hooke in 1665**.

→ DNA carries a **negative charge** due to phosphate groups.

→ Mixture of DNA and proteins is known as **chromatin**.

→ Cyclin D and Cyclin E are produced during the cell cycle's **G1 phase.**

→ Lipids with double bonds are **liquid (unsaturated fats).**

→ UUU codes for the **amino acid phenylalanine**.

→ An animal cell contains around ***10 billion*** protein molecules.

→ Chromosomes' translational value ranges from **genes to proteins.**

→ Helical structures that bind against **histones**.

→ Protein activators consist of **multiple domains**.

Zoology Legends

→ The human body has approximately **20 million** protein-coding genes.

→ Intrinsic protein is a **membrane-spanning protein**.

→ Inhibitors that bind to the active site of enzymes are **Competitive Inhibitors.**

→ Inhibitors that bind to the non-active site of enzymes are **Non-competitive Inhibitors.**

→ Beta pleated sheets are found in **Secondary Proteins**.

→ An atom can form a maximum of **4 covalent bonds**.

→ Inhibitors that bind to a site other than the active site of an enzyme **are Irreversible Inhibitors**.

→ ABO blood group is the best example of **Co-dominance**.

→ Starch and glycogen are stored forms of ***glucose in Animals***, while cellulose is a stored **glucose form in Plants**.

→ Different forms of the same gene are **called RFLPs (Restriction Fragment Length Polymorphisms).**

→ **Frederick Griffith** is known for discovering that **DNA** is the main hereditary molecule transferred between generations.

→ Missing enzyme homogenetic acid oxidase causes **Phenylketonuria (PKU).**

→ Beta pleated sheets in proteins are joined by **Hydrogen Bonds**.

→ Mutation that results in no change in the amino acid is a **Silent Mutation.**

→ Calmodulin is about **10%** of cell protein mass.

Zoology Legends

→ The protein that stimulates breast cells to produce milk is **Prolactin**.

→ The initiation codon is **AUG**.

→ TH2 cells mainly help to activate **B cells (B lymphocytes).**

→ Mendel denoted dominant and recessive traits as **SS and ss.**

→ For optimal activity of hydrolytic enzymes, lysosomes maintain a pH around **5.0.**

→ Proteins fully translocated across the Endoplasmic Reticulum and embedded in the membrane are **Integral Membrane Proteins.**

→ Ran GTPase is a molecular switch that exists in **two states: GTP-bound and GDP-bound.**

→ An oncogene after mutation can produce an abnormal protein that contributes to **cancer**.

→ Dendritic cells use cytokines such as **IL-2 to** guide helper T cells.

→ An average cell divides about **50 times** before it becomes unhealable (senescent).

→ Cholera toxin prevents hydrolysis of **GTP to GDP**, leading to a prolonged G-protein signaling.

→ The immune system, like **the nervous system**, can remember prior experiences through **Adaptive immunity**.

→ There are about **2 x 10^12 lymphocytes** in the human body.

→ Cells synthesizing DNA need to import about **10^6 histone** molecules from the cytosol every **3 minutes.**

Zoology Legends

→ An animal cell contains about **10^6 proteins**.

→ The total amount of lymphocytes in the human body is approximately **2 x 10^12**.

→ A cluster of genes with one promoter is called an **operon**.

→ Binding of ribosomal subunits to the start codon is called the **Shine-Dalgarno sequence**.

→ Undifferentiated cells of higher organisms capable of division are **stem cells.**

→ In almost **30%** of cancer tumors in humans, the RAS gene is mutated.

→ Optimal activity of hydrolytic enzymes in lysosomes is maintained at a pH of around **5.0**.

→ **EDTA** is used as a **cryoprotectant**.

→ Calmodulin is about **10%** of cell protein mass.

→ Cholera toxin prevents the hydrolysis of **GTP to GDP**.

→ **Autocrine signals** bind to receptors on the same cells that secrete them.

→ Xist is the gene on the **X chromosome** that inactivates it.

→ Adopter proteins are involved in moving signals from **cytosol to the nucleus.**

→ Ribosomes are engaged to **mRNA** during translation.

→ The **adaptive** immune system can remember prior experiences, similar to the **nervous system**.

Zoology Legends

→ There are **about 2 x 10^12** lymphocytes in the human body.

→ **Introns** are removed during RNA maturation.

→ Activators consist of **multiple** domains.

→ DNA carries a **negative** charge.

→ A mixture of DNA and proteins forms **chromatin**.

→ Protein mutation involves changes in **base pairs**.

→ The translational value range of chromosomes varies based on the ***specific genes***.

→ Smad complex moves towards the **nucleus**.

→ The approximate number of lymphocytes in the human body is **2 x 10^12**.

→ Autonomous cell growth is present in **cancer**.

→ Philadelphia chromosome is a chromosomal translocation between chromosomes **9 and 22**.

→ There are **five classes** of antibodies.

→ Ran GTPase is a molecular switch that exists in **two states.**

→ TH2 cells mainly help activate **B cells.**

→ The discovery of blood group systems is attributed to various scientists, including ***Karl Landsteiner***.

→ The approximate origin of life is around 4 billion years ago.

→ The two essential sequences present in the promoter region are the **TATA box and transcription start site**.

→ Kidney transplant was first performed by **Joseph Murray.**

Zoology Legends

→ Actin stabilizes **cell shape**.

→ Undifferentiated cells are called **stem cells.**

→ The two promoter regions are the **TATA box and recognition site**.

→ Ran GTPase is a molecular switch that exists in **two states**.

→ Antagonists **inhibit** hormone production.

→ Large subunits of ribosomes contain polypeptides **45 polypeptides**

→ An adapter molecule possesses **two distinct regions** - one region binds to a specific amino acid, while another region recognizes a sequence of nucleotides

→ **Restriction endonucleases** (restriction enzymes) act as a kind of immune system, protecting the cell from the invasion of foreign DNA (virus).

→ Phosphorylation signaling in the E. coli alters__***Protein Synthesis***

→ Calmodulin binds 4 Ca 2+ allosteric (PKA) ----? **1%**

→ Enzyme that can cut DNA at specific site____**Restriction enzymes**

→ Protein folding information is a function of **AA** sequence.

→ Structure formed by many folding of polypeptide chain___**Secondary protein structure**

→ IgG binds Polyvalent **100 times** stronger than monovalent

Zoology Legends

→ True Arrangement step of mitosis.....>**Interphase>Prophase>Premetaphase>Metaphase >Anaphase>Telophase**

→ Which inhibitor inactivate the enzyme activity ..... **irreversible inhibitor**

→ Haemocytometer measure only **blood cells**.

→ Some cancer cells acquire the ability through mutations to produce growth factors to which they are responsive **(autocrine signaling)**.

→ Which signal receptors on the cells that secrete them? **Autocrine signal**

→ Polar R chain Amino acid...**Hydrophilic**

→ Autocrine signals bind to the receptors on the cells, which..... **Cells that secrete them**

→ Mitochondria kdr sy ata ha(mother, father ,both ,none)

→ Part of **promoter** is initiation site **(transcription start).**

→ T.H morgan dicovered__? **Gene linkage**

→ Maltose has___(**glucose + glucose alpha 1,4 glycosidic linkage)**

→ Filters come in two pore sizes **0.2 um and 0.45** um

→ **0.45** is primarily used to remove particulates from solutions

→ Polypeptide chain has _(alpha chains, beta chains, both ,none)

→ failure of sister chromatids to separate **(second meiotic division**

Zoology Legends

→ Laminar flow Provides **sterile working area** where cells can be handled without worrying about contamination.

→ Rate of threshold effect in identical twins__**20-40% and 2%-7%** in for first degree relatives

→ Phosphorylation signaling in E.Coli alter?_**Protein Synthesis**

→ In which stage of cell division the size of chromosomes is best seen?_**Metaphase**

→ **Metaphase** is the best time to see the sizes and shapes of chromosomes because they are maximally condensed.

→ **Single bonds** .....between the carbon atoms.

→ Prime Ends .....**(3'& 5')**

→ **The Miracle Workers:** One of biggest miracles of life is that it can perform slow reactions and the ones requiring extreme conditions, quickly and at physiological conditions.

→ Billions of cells die each day in humans, **(blood epithelia)**

→ **Epistasis:** Effect of one gene dependent on the presence of one or more 'modifier genes'

→ Protein active during DNA damage Histone, actin , kinase

→ mendal second law__**Law of independent**

→ sex influence **(X linked)**

→ replication occur in (**S phase**)

→ microorganisms that are pass through **0.2 micro m** filter **(Mycoplasma and viruses)**

Zoology Legends

→ operon is related with (**promoter**)

→ cause of down syndrome **(Maternity age)**

→ Down Syndrome: Arises from **meiotic** **nondisjunction**

→ first Ig that reach first against pathogen **(lgM)**

→ transcription produced **(Single stranded RNA)**

→ Mendal ny konse plant py work Kiya tha? **Pea plants**

→ one is not the function of lipids?

→ A palindromic sequence is the same when read in **5′ to 3′** direction on either complementary strand of DNA.

→ Meiosis divided into _ cells? **Four**

→ Which cells release histamine when damaged? **Mast cells**

→ T cells mature in ? ***Thymus***

→ After translation, ----can be removed? ***Methionine***

→ Beta blockers are involved in .....**mimicking the function of a hormone**.

→ Initiation site of transcription--- **Promoter**

→ Helices that are involved in against binding are...**Helix5**

→ Function of Golgi apparatus is ------ **Packing and transportation of proteins**

→ An average globular cell has-----mitochondria. **10,000**

→ TH2 cells mainly help to activate… **B cells**

→ Which is the first class of Ab to appear on the cells surface? ***lgM***

Zoology Legends

→ Oncogenes after mutation produce one of abnormal condition……
Cancer

→ Inactive ICR may be DNA bound or **in cytoplasm** (NLS nonfunctional)

→ **Apoptotic cells** lose the asymmetric distribution of phospholipids in their plasma membrane as consequence, negatively charged **phosphatidylserine**,

→ Type of IG reach first at site of infection…..**IgM**

→ DMSO and glycerol is example of …**Cryoprotectants**

→ **Competitive inhibiters** are similar to the natural substrate (bind the **active site)**

→ **Collagen** provides strength to mammalian skin.

→ **Gregor Mendel** is the father of genetics.

→ Golgi apparatus functions in **protein modification and sorting**.

→ Protein has a **3-dimensional** structure.

→ The larger ribosomal subunit contains **3 RNAs**.

→ Phosphorylation in E. coli alters **protein synthesis**.

→ IGF-1 promotes **cell growth and proliferation**.

→ DNA replication occurs in **the S phase**.

→ Four **macromolecules:** carbohydrates, proteins, lipids, nucleic acids.

→ Mendel's laws determine **inheritance probability**.

→ Formation of RNA is **called transcription**.

Zoology Legends

→ Translation occurs in **ribosomes**

Repeated Questions Spring 2023

- → **Function of TH1 and TH2 cells**
- → **Function of ERGICf3**
- → **Types of MHC molecules**
- → **Difference between endocytosis and exocytosis**
- → **Differences between mitosis and meiosis**
- → **Differences between B cells and T cells**
- → **Role of prokaryotic organisms in living organisms**
- → **Function of endoplasmic reticulum**
- → **Activation of T cells**
- → **Function of the immune system**
- → **Function of mitochondria**
- → **Differences between ERGICf3 and ERGICt cells**
- → **Functions of endoplasmic reticulum cytotrans and endoplasmic reticulum helper T cells**
- → **Function of autophagy**
- → **Definition of rod cells**
- → **Differences between DNA and RNA**
- → **Differences between confluent monolayer and post-confluent cells**

Zoology Legends

- → Differences between homotypic and heterotypic fusion
- → Strategies of transcription regulation
- → Clonal selection process
- → Transcription process
- → Differences between MHC class I and MHC class II molecules
- → Definition of differentiation between autosomes and meiosis
- → Function of protein
- → Subtypes of IgG
- → Differences between humoral immunity and cellular immunity
- → Function of cAMP in structural movement
- → Function of protein ERGICf3
- → Function of protein ERGICt3
- → Function of endocytosis and exocytosis in cell transport
- → Role of regulatory region and explain two elements.
- → Features of adaptive immune system.
- → Five subclasses of human IgG.
- → Difference between multiple-pass transmembrane protein and double-pass transmembrane protein.
- → Function of ERGIC53 protein.
- → RNA processing steps.
- → Cellular immunity.

Zoology Legends

- → Laminar flow.
- → Hemocytometer.
- → UV sterilization equipment.
- → Genotype, phenotype, homozygous, heterozygous.
- → Requirements for PCR.
- → Five classes of antibodies.
- → Components of the cell membrane.
- → Function of Golgi complex.
- → Difference between cytotoxic and helper T cells.
- → Function of PI3K (Phosphoinositide 3-kinase).
- → Explain cell cycle.
- → How do β-blockers affect the function of adrenaline?
- → How does UV sterilization help with equipment sterilization?
- → Explain the mechanism of growth promotion by oncogenes.
- → Applications of PCR.
- → Adult stem cells: definition, benefits, and therapeutic uses.
- → Explain the interphase and its phases.
- → Describe the subclasses of IgG.
- → Describe the synapse and its role at the neuromuscular junction.
- → Explain the lecithin pathway and complementary regulation.
- → Write a note on genetic engineering.

Zoology Legends

→ Explain the laminar flow cabinet principle and function.
→ Describe the rod cells.
→ Write a note on clonal selection.
→ Explain the transcription process.
→ Explain the Latin square experiment.
→ Explain the autonomous cell growth factors.
→ Describe the mechanism of action of β-blockers on adrenaline.
→ Explain the function of Golgi complex.
→ Name three RNA viruses.
→ Explain the properties of the immune system.
→ Explain MHC and its functions.
→ Explain how UV light disinfection works.
→ Explain the importance of the cell.
→ Explain the role of CAM kinase II in muscle movement.
→ Function of plasma membrane.
→ Functions of intracellular receptors.
→ Five names of growth oncogenes.
→ Importance of cells.
→ N-terminal signaling and its function in the endoplasmic reticulum.
→ Structure of an antibody.
→ Gel electrophoresis.
→ Mechanism of T cell activation.

Zoology Legends

- → Semiconservative, conservative, and dispersive replication of DNA.
- → Adult stem cells, benefits, and therapeutic uses.
- → Function of DNA.
- → DNA packaging inside the cell.
- → Activation of hypertoxic T cells.
- → Types of cristae.
- → Helper T cells and cytotoxic T cells.
- → Strategies of transcription.
- → Haemocytometer.
- → N-terminal signaling and its function (3 marks).
- → Immune system (3 marks).
- → Adult stem cells, their benefits, and therapeutics (5 marks).
- → Rod cells (3 marks).
- → Antibiotics (3 marks).
- → Regulatory proteins (5 marks).
- → Application of polymerase chain reaction (5 marks).
- → Transcription (5 marks).
- → Synapse in the neuromuscular system.
- → Cryoprotectant and examples.
- → Endocytosis, phagocytosis, pinocytosis, and exocytosis.
- → Lectin pathway complement system regulation.
- → Properties of the immune system.
- → Functions of the endoplasmic reticulum.

Zoology Legends

→ Three steps of RNA processing.
→ Clonal selection.
→ Sterilizing agents (five names).
→ Role of lipids.
→ Subclasses of human IgG.
→ Signaling functions of the N-terminal ER signal sequence.
→ Structures of antibodies.
→ Signaling in E. coli.
→ How proteins are transported in mitochondria.
→ Difference between DNA and RNA.
→ Description of the immune system.
→ Function of N-terminal signaling.
→ Transcription strategies.
→ Components of the cellular membrane.
→ Endocytosis, phagocytosis, and pinocytosis.
→ Primary culture, cell line, and pure culture.
→ Interphase and its three stages.
→ E. coli signaling.
→ N signaling in cells.
→ Clonal selection.
→ Conservative, semiconservative models of replication of DNA.
→ Regulatory proteins and their elements.
→ Promotion of cancer in cells by certain elements.
→ MHC and its types.

Zoology Legends

→ UV light sterilization of equipment.
→ Regulatory region and its elements.
→ Role of lipid bilayer.
→ Homo vs. heterotypic fusion.
→ Note on transcription.

Muhammad Zaman

03044756496

Remember me in you Precious Prayer

Jazak Allah

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